

EXP-2

Reading data / some functions

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```
import pandas as pd
import numpy as np
```

```
# =====
# 1. CONCATENATION OF DATAFRAMES
# =====
```

```
dataFrame1 = pd.DataFrame({
'StudentID': [1, 3, 5, 7, 9],
'Score': [89, 39, 50, 97, 22]
})
```

```
dataFrame2 = pd.DataFrame({
'StudentID': [2, 4, 6, 8, 10],
'Score': [98, 93, 44, 77, 69]
})
```

```
# Vertical Concatenation
vertical_concat = pd.concat([dataFrame1, dataFrame2],
ignore_index=True)
print("Vertical Concatenation:")
print(vertical_concat)
```

```
# Horizontal Concatenation
```

```
horizontal_concat = pd.concat([dataFrame1, dataFrame2],  
axis=1)  
print("\nHorizontal Concatenation:")  
print(horizontal_concat)
```

```
# =====  
# 2. MERGING DATAFRAMES  
# =====
```

```
dfSE = pd.DataFrame({  
    'StudentID': [2, 4, 6, 8, 10, 12],  
    'ScoreSE': [98, 93, 44, 77, 69, 56]  
})
```

```
dfML = pd.DataFrame({  
    'StudentID': [1, 2, 3, 4, 5, 6, 8, 10],  
    'ScoreML': [39, 93, 49, 44, 55, 78, 97, 87]  
})
```

```
# Inner Join  
inner_join = pd.merge(dfSE, dfML, on='StudentID', how='inner')  
print("\nInner Join:")  
print(inner_join)
```

```
# Left Join  
left_join = pd.merge(dfSE, dfML, on='StudentID', how='left')  
print("\nLeft Join:")  
print(left_join)
```

```
# Right Join
```

```
right_join = pd.merge(dfSE, dfML, on='StudentID', how='right')
print("\nRight Join:")
```

```
print(right_join)
```

```
# Outer Join
```

```
outer_join = pd.merge(dfSE, dfML, on='StudentID', how='outer')
print("\nOuter Join:")
print(outer_join)
```

```
# =====
```

```
# 3. ADDING A NEW COLUMN
```

```
# =====
```

```
sales = pd.DataFrame({
    'Quantity': [100, 250, 300, 150],
    'UnitPrice': [50, 60, 45, 80]
})
```

```
sales['TotalPrice'] = sales['Quantity'] * sales['UnitPrice']
```

```
print("\nSales Data:")
print(sales)
```

```
# =====
```

```
# 4. DESCRIPTIVE STATISTICS
```

```
# =====
```

```
print("\nDescriptive Statistics:")
print(sales.describe())
```

```
# =====  
# 5. VALUE COUNTS  
# =====  
company_df = pd.DataFrame({  
    'Company': ['ABC', 'XYZ', 'ABC', 'PQR', 'XYZ', 'ABC']  
})  
print("\nCompany Counts:")  
print(company_df['Company'].value_counts())
```